



Math 4543: Numerical Methods

Lecture 13 — Simpson's $1/3$ Rule and Simpson's $3/8$ Rule of Integration

Syed Rifat Raiyan

Lecturer

Department of Computer Science & Engineering
Islamic University of Technology, Dhaka, Bangladesh

Email: rifatraiyan@iut-dhaka.edu

Lecture Plan

The agenda for today

- Understand the idea behind the Simpson's rules of integration
- Know about the derivation approaches of single-segment Simpson's $1/3$ rule formula †
- Derive the Multiple-segment Simpson's $1/3$ rule formula
- Know about the derivation approaches of single-segment Simpson's $3/8$ rule formula †
- Derive the Multiple-segment Simpson's $3/8$ rule formula
- Understand how the $1/3$ and $3/8$ rules can be fused or mixed together

† means that the topic is not important for exams.

Simpson's Rules

What are they?

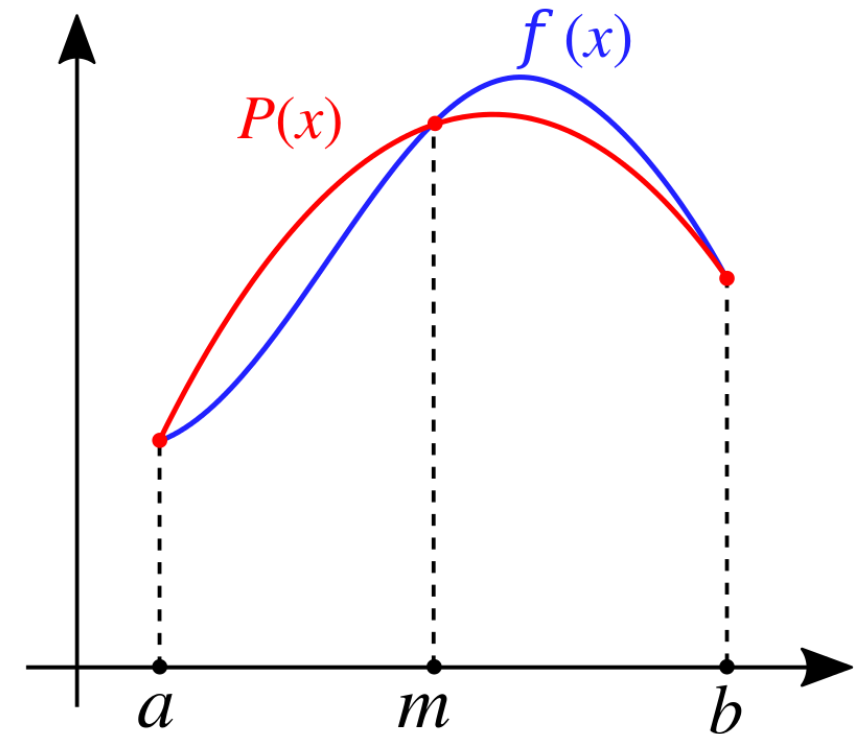
Simpson's rules are methods of *numerical integration* invented by mathematician Thomas Simpson.

The idea is to *approximate the definite integral* of a function $f(x)$ by calculating the definite integral of an n^{th} order polynomial $f_n(x) = P(x)$ that *approximates the function* $f(x)$. [$n > 1$]

We'll mainly focus on **two** Simpson's rules —

- Simpson's 1/3 Rule
- Simpson's 3/8 Rule

We will also solve an example using *both of these rules* together.



Simpson's 1/3 Rule

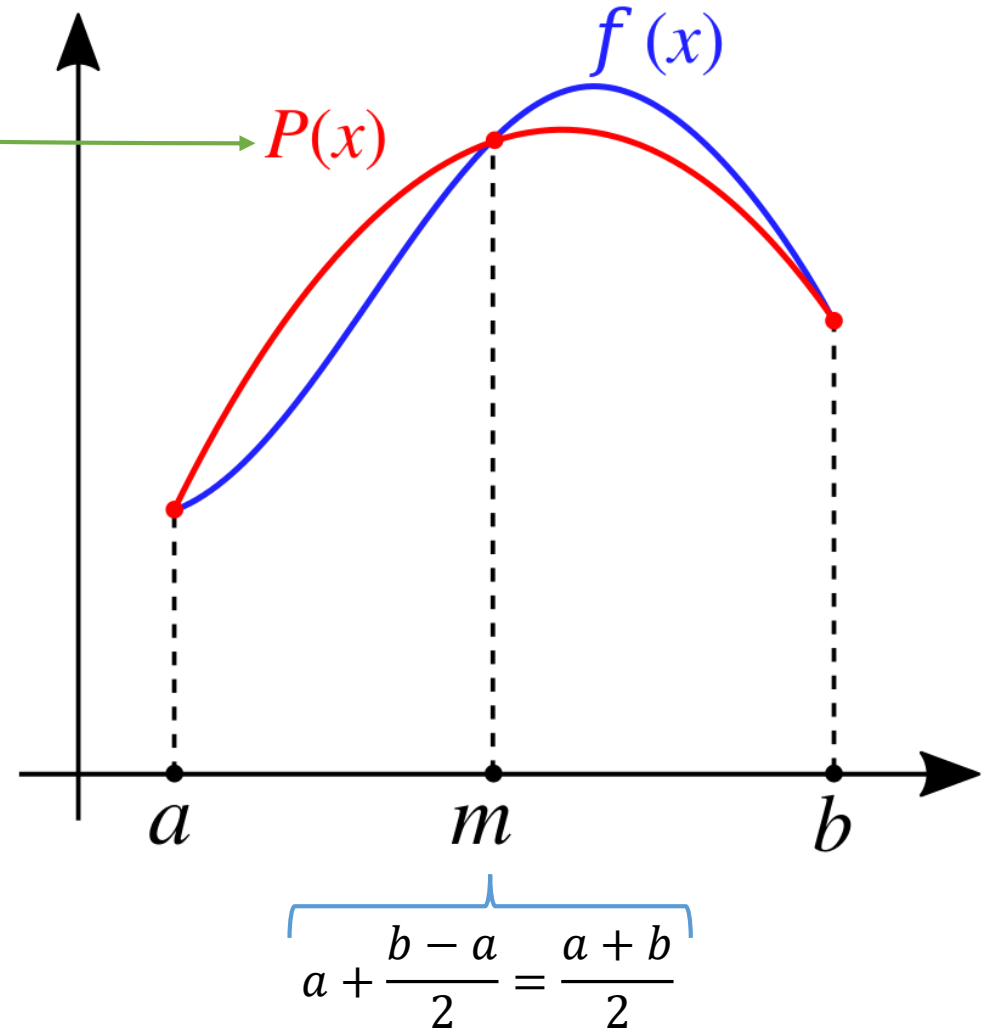
The general idea

Simpson's 1/3 rule is an extension of Trapezoidal rule where the integrand is approximated by a *second order* polynomial $f_2(x)$.

$$I = \int_a^b f(x) dx \approx \int_a^b f_2(x) dx$$

where $f_2(x)$ is a second order polynomial given by

$$f_2(x) = a_0 + a_1x + a_2x^2$$



Simpson's 1/3 Rule

Deriving the formula †

Choose

$$(a, f(a)), \left(\frac{a+b}{2}, f\left(\frac{a+b}{2}\right) \right), \text{ and } (b, f(b))$$

as the three points of the function to evaluate a_1 and a_2 .

$$f(a) = f_2(a) = a_0 + a_1a + a_2a^2$$

$$f\left(\frac{a+b}{2}\right) = f_2\left(\frac{a+b}{2}\right) = a_0 + a_1\left(\frac{a+b}{2}\right) + a_2\left(\frac{a+b}{2}\right)^2$$

$$f(b) = f_2(b) = a_0 + a_1b + a_2b^2$$

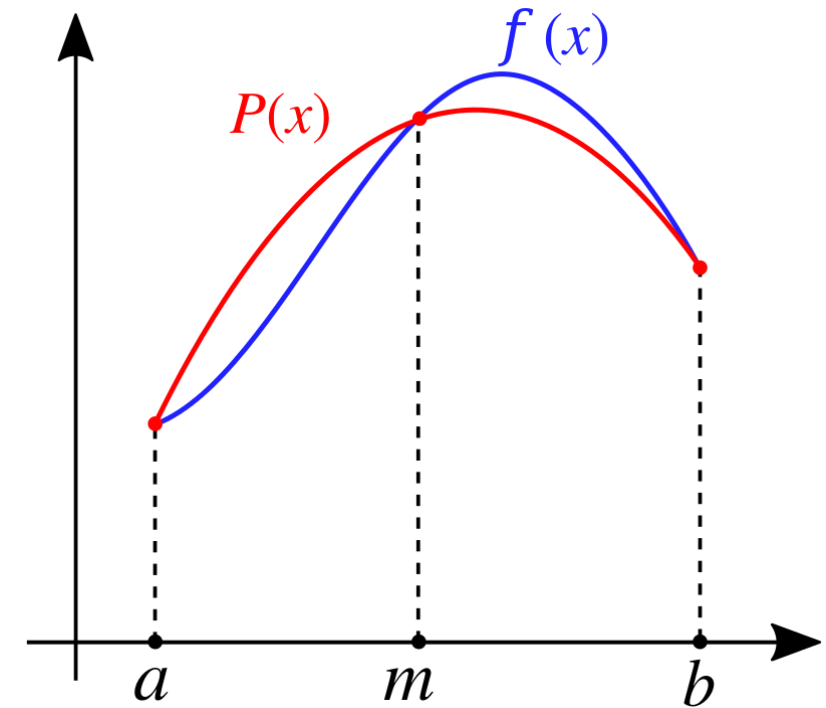
Solving the above three equations for unknowns, a_0 , a_1 and a_2 give

$$a_0 = \frac{a^2f(b) + abf(b) - 4abf\left(\frac{a+b}{2}\right) + abf(a) + b^2f(a)}{a^2 - 2ab + b^2}$$

Using symbolic solver software *e.g.* MATLAB, Octave, Maple, WolframAlpha etc.

$$a_1 = -\frac{af(a) - 4af\left(\frac{a+b}{2}\right) + 3af(b) + 3bf(a) - 4bf\left(\frac{a+b}{2}\right) + bf(b)}{a^2 - 2ab + b^2}$$

$$a_2 = \frac{2\left(f(a) - 2f\left(\frac{a+b}{2}\right) + f(b)\right)}{a^2 - 2ab + b^2}$$



$$a + \frac{b-a}{2} = \frac{a+b}{2}$$

Simpson's 1/3 Rule

Deriving the formula †

Then

$$\begin{aligned} I &\approx \int_a^b f_2(x) dx \\ &= \int_a^b (a_0 + a_1x + a_2x^2) dx \\ &= \left[a_0x + a_1\frac{x^2}{2} + a_2\frac{x^3}{3} \right]_a^b \\ &= a_0(b-a) + a_1\frac{b^2-a^2}{2} + a_2\frac{b^3-a^3}{3} \end{aligned}$$

Substituting values of a_0 , a_1 and a_2 give

$$\int_a^b f_2(x) dx = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Since for the single application of Simpson 1/3 rule, the interval $[a, b]$ is broken into 2 segments, the segment width

$$h = \frac{b-a}{2}$$

The Simpson's 1/3 rule can be rewritten as

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Since the above form has 1/3 in its formula, it is called Simpson's 1/3 rule.

Go through the lecture note PDF to know about the other approaches of deriving the same formula.

Simpson's 1/3 Rule

An example

The distance covered by a rocket in meters from $t = 8\text{s}$ to $t = 30\text{s}$ is given by

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use Simpson's 1/3 rule to find the approximate value of x .
- b) Find the true error, E_t .
- c) Find the absolute relative true error, $|\epsilon_t|$.

Simpson's 1/3 Rule

An example

Solution

$$a) \quad x \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$a = 8$$

$$b = 30$$

$$\frac{a+b}{2} = 19$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$f(8) = 2000 \ln \left[\frac{140000}{140000 - 2100(8)} \right] - 9.8(8) = 177.27m/s$$

$$f(30) = 2000 \ln \left[\frac{140000}{140000 - 2100(30)} \right] - 9.8(30) = 901.67m/s$$

$$f(19) = 2000 \ln \left(\frac{140000}{140000 - 2100(19)} \right) - 9.8(19) = 484.75m/s$$

$$\begin{aligned} x &\approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] \\ &= \left(\frac{30-8}{6} \right) [f(8) + 4f(19) + f(30)] \\ &= \frac{22}{6} [177.27 + 4 \times 484.75 + 901.67] \\ &= 11065.72m \end{aligned}$$

b) The exact value of the above integral is

$$\begin{aligned} x &= \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\ &= 11061.34m \end{aligned}$$

$$\begin{aligned} \text{So the true error is } E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061.34 - 11065.72 = -4.38m \end{aligned}$$

c) The absolute relative true error is

$$\begin{aligned} |\epsilon_t| &= \left| \frac{\text{True Error}}{\text{True Value}} \right| \times 100 \\ &= \left| \frac{-4.38}{11061.34} \right| \times 100 \\ &= 0.0396\% \end{aligned}$$

Multi-segment Simpson's 1/3 Rule

The composite version

Just like in composite trapezoidal rule, one can subdivide the interval $[a, b]$ into n segments and apply Simpson's 1/3 rule repeatedly over every two segments. Note that n needs to be even. Divide interval $[a, b]$ into n equal segments, so that the segment width is given by

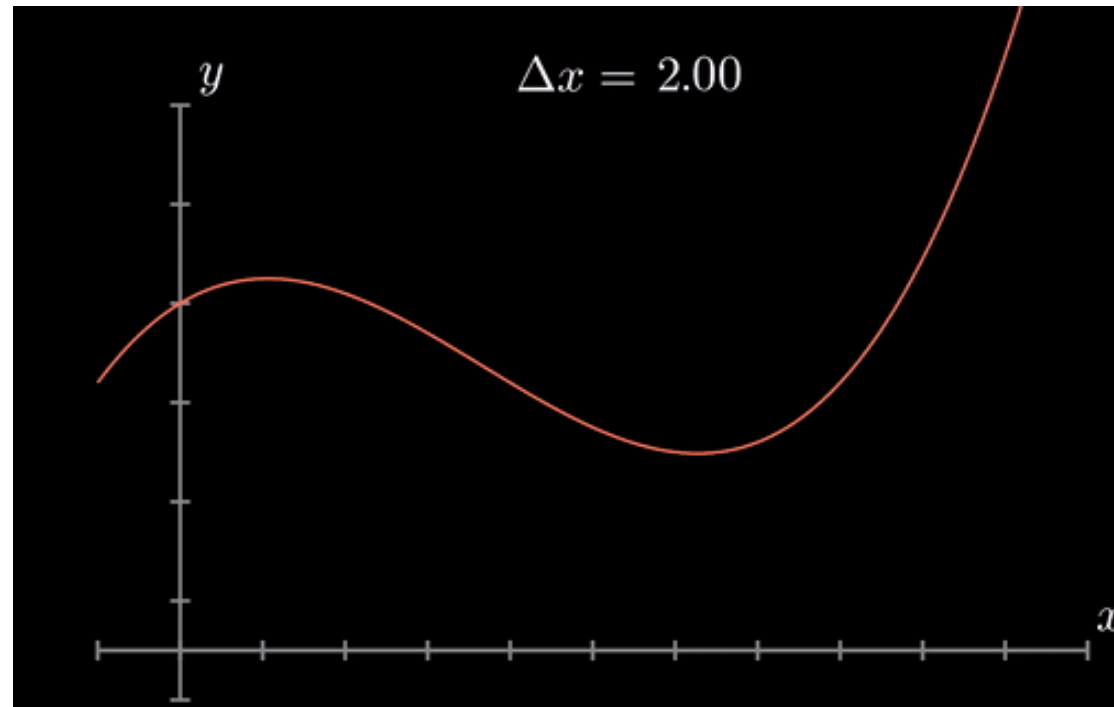
$$h = \frac{b - a}{n}$$

Now

$$\int_a^b f(x)dx = \int_{x_0}^{x_n} f(x)dx$$

where $x_0 = a$ $x_n = b$

$$\int_a^b f(x)dx = \int_{x_0}^{x_2} f(x)dx + \int_{x_2}^{x_4} f(x)dx + \dots + \int_{x_{n-4}}^{x_{n-2}} f(x)dx + \int_{x_{n-2}}^{x_n} f(x)dx$$



Adapted from
www.youtube.com/watch?v=DdNAcvrezc

Multi-segment Simpson's 1/3 Rule

Deriving the formula

$$\int_a^b f(x)dx = \int_{x_0}^{x_2} f(x)dx + \int_{x_2}^{x_4} f(x)dx + \dots + \int_{x_{n-4}}^{x_{n-2}} f(x)dx + \int_{x_{n-2}}^{x_n} f(x)dx$$

Apply Simpson's 1/3rd Rule over each interval,

$$\begin{aligned} \int_a^b f(x)dx \cong & (x_2 - x_0) \left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{6} \right] + (x_4 - x_2) \left[\frac{f(x_2) + 4f(x_3) + f(x_4)}{6} \right] + \dots + \\ & (x_{n-2} - x_{n-4}) \left[\frac{f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})}{6} \right] + (x_n - x_{n-2}) \left[\frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6} \right] \end{aligned}$$

Since

$$\begin{aligned} x_i - x_{i-2} = 2h \\ i = 2, 4, \dots, n \end{aligned} \quad \text{then} \quad \int_a^b f(x)dx \cong 2h \left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{6} \right] + 2h \left[\frac{f(x_2) + 4f(x_3) + f(x_4)}{6} \right] + \dots$$
$$+ 2h \left[\frac{f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})}{6} \right] + 2h \left[\frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6} \right]$$

$$= \frac{h}{3} [f(x_0) + 4(f(x_1) + f(x_3) + \dots + f(x_{n-1})) + 2(f(x_2) + f(x_4) + \dots + f(x_{n-2})) + f(x_n)]$$

$$= \frac{h}{3} \left[f(x_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n) \right] \cong \frac{b-a}{3n} \left[f(x_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n) \right]$$

Multi-segment Simpson's 1/3 Rule

The same example (now with *multiple segments*)

Use composite Simpson's 1/3 rule with 4 segments to approximate the distance covered by a rocket in meters from $t = 8\text{s}$ to $t = 30\text{s}$ as given by

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use composite Simpson's 1/3rd rule with 4 segments to estimate x .
- b) Find the true error, E_t for part (a).
- c) Find the absolute relative true error, $|\epsilon_t|$ for part (a).

Multi-segment Simpson's 1/3 Rule

The same example (now with *multiple segments*)

Solution:

a) Using composite Simpson's 1/3 rule,

$$x \approx \frac{b-a}{3n} \left[f(t_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(t_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(t_i) + f(t_n) \right]$$

$$\begin{aligned} n &= 4 & h &= \frac{b-a}{n} \\ a &= 8 & &= \frac{30-8}{4} \\ b &= 30 & &= 5.5 \end{aligned}$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

So $f(t_0) = f(8)$

$$\begin{aligned} f(8) &= 2000 \ln \left[\frac{140000}{140000 - 2100(8)} \right] - 9.8(8) \\ &= 177.27m/s \end{aligned}$$

$$f(t_1) = f(8 + 5.5) = f(13.5)$$

$$\begin{aligned} f(13.5) &= 2000 \ln \left[\frac{140000}{140000 - 2100(13.5)} \right] - 9.8(13.5) \\ &= 320.25m/s \end{aligned}$$

$$f(t_2) = f(13.5 + 5.5) = f(19)$$

$$\begin{aligned} f(19) &= 2000 \ln \left(\frac{140000}{140000 - 2100(19)} \right) - 9.8(19) \\ &= 484.75m/s \end{aligned}$$

$$f(t_3) = f(19 + 5.5) = f(24.5)$$

$$\begin{aligned} f(24.5) &= 2000 \ln \left[\frac{140000}{140000 - 2100(24.5)} \right] - 9.8(24.5) \\ &= 676.05m/s \end{aligned}$$

$$f(t_4) = f(t_n) = f(30)$$

$$\begin{aligned} f(30) &= 2000 \ln \left[\frac{140000}{140000 - 2100(30)} \right] - 9.8(30) \\ &= 901.67m/s \end{aligned}$$

Multi-segment Simpson's 1/3 Rule

The same example (now with *multiple segments*)

$$\begin{aligned}x &= \frac{b-a}{3n} \left[f(t_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(t_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(t_i) + f(t_n) \right] \\&= \frac{30-8}{3(4)} \left[f(8) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^3 f(t_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^2 f(t_i) + f(30) \right] \\&= \frac{22}{12} [f(8) + 4f(t_1) + 4f(t_3) + 2f(t_2) + f(30)] \\&= \frac{11}{6} [f(8) + 4f(13.5) + 4f(24.5) + 2f(19) + f(30)] \\&= \frac{11}{6} [177.27 + 4(320.25) + 4(676.05) + 2(484.75) + 901.67] \\&= 11061.64m\end{aligned}$$

b) The exact value of the above integral is

$$\begin{aligned}x &= \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\&= 11061.34m\end{aligned}$$

So the true error is

$$\begin{aligned}E_t &= \text{True Value} - \text{Approximate Value} \\E_t &= 11061.34 - 11061.64 \\&= -0.30m\end{aligned}$$

c) The absolute relative true error is

$$\begin{aligned}|\epsilon_t| &= \left| \frac{\text{True Error}}{\text{True Value}} \right| \times 100 \\&= \left| \frac{-0.3}{11061.34} \right| \times 100 \\&= 0.0027\%\end{aligned}$$

Multi-segment Simpson's 1/3 Rule

The same example (now with *multiple segments*)

Table 1 Values of composite Simpson's 1/3 rule for Example 1

n	Approximate Value	E_t	$ \epsilon_t $
2	11065.72	-4.38	0.0396%
4	11061.64	-0.30	0.0027%
6	11061.40	-0.06	0.0005%
8	11061.35	-0.02	0.0002%
10	11061.34	-0.01	0.0001%

The trend is that the *error values consistently decrease*
the *more we increase the number of segments*.

Simpson's 3/8 Rule

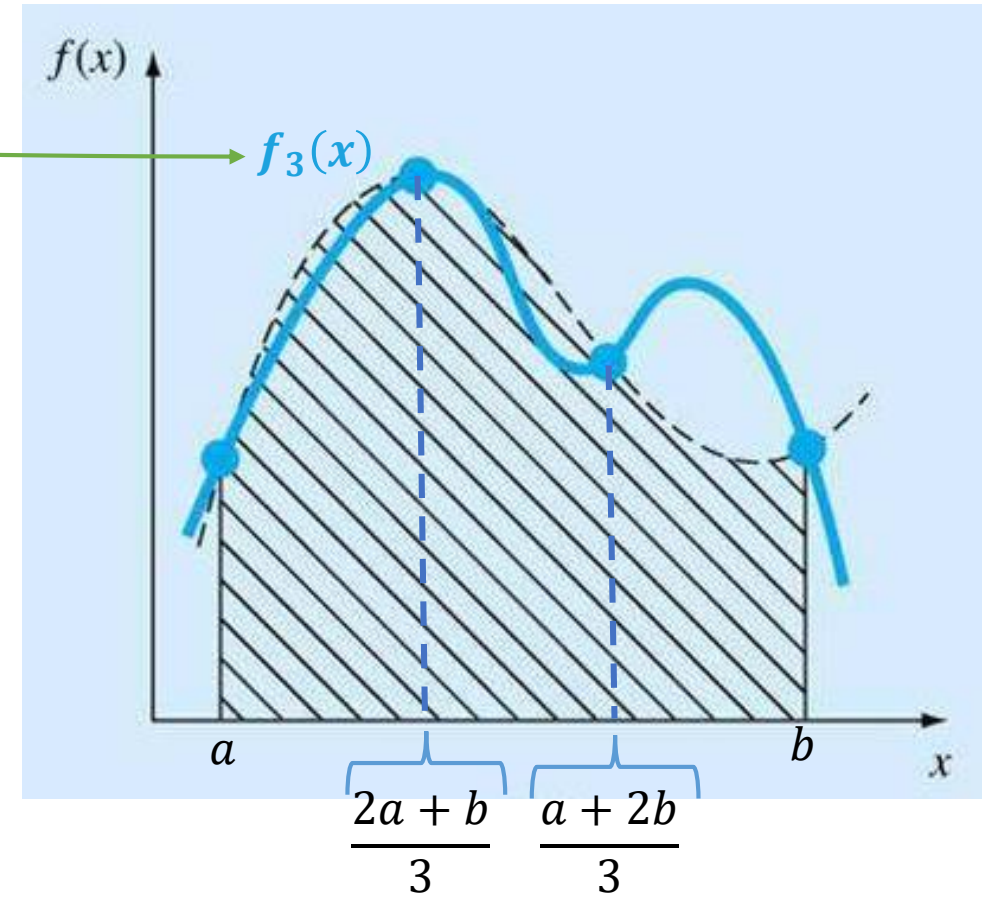
The general idea

Simpson's 3/8 rule is a numerical integration method where the integrand is approximated by a *third order* polynomial $f_3(x)$.

$$I = \int_a^b f(x) dx \approx \int_a^b f_3(x) dx \quad (1)$$

where $f_3(x)$ is a 3rd order polynomial given by

$$f_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3 \quad (2)$$



Simpson's 3/8 Rule

Deriving the formula †

In a similar fashion, Simpson 3/8 rule for integration can be derived by approximating the given function $f(x)$ with the 3rd order (cubic) polynomial $f_3(x)$

$$\left. \begin{aligned} f_3(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 \\ &= \{1, x, x^2, x^3\} \times \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} \end{aligned} \right\} \quad (3)$$

The unknown coefficients a_0 , a_1 , a_2 and a_3 in Equation (3) can be obtained by substituting 4 known coordinate data points $\{x_0, f(x_0)\}$, $\{x_1, f(x_1)\}$, $\{x_2, f(x_2)\}$ and $\{x_3, f(x_3)\}$ into Equation (3) as follows.

$$\left. \begin{aligned} f(x_0) &= a_0 + a_1x_0 + a_2x_0^2 + a_3x_0^3 \\ f(x_1) &= a_0 + a_1x_1 + a_2x_1^2 + a_3x_1^3 \\ f(x_2) &= a_0 + a_1x_2 + a_2x_2^2 + a_3x_2^3 \\ f(x_3) &= a_0 + a_1x_3 + a_2x_3^2 + a_3x_3^3 \end{aligned} \right\} \quad (4)$$

Equation (4) can be expressed in matrix notation as

$$\begin{bmatrix} 1 & x_0 & x_0^2 & x_0^3 \\ 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ 1 & x_3 & x_3^2 & x_3^3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ f(x_3) \end{bmatrix} \quad (5)$$

The above Equation (5) can symbolically be represented as

$$[A]_{4 \times 4} \vec{a}_{4 \times 1} = \vec{f}_{4 \times 1} \quad (6)$$

The above Equation (5) can symbolically be represented as

$$[A]_{4 \times 4} \vec{a}_{4 \times 1} = \vec{f}_{4 \times 1} \quad (6)$$

Thus,

$$\vec{a} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = [A]^{-1} \times \vec{f} \quad (7)$$

Simpson's 3/8 Rule

Deriving the formula †

Substituting Equation (7) into Equation (3), one gets

$$f_3(x) = \{1, x, x^2, x^3\} \times [A]^{-1} \times \vec{f} \quad (8)$$

$$x_0 = a$$

$$x_1 = a + h$$

$$= a + \frac{b-a}{3}$$

$$= \frac{2a+b}{3}$$

$$x_2 = a + 2h$$

$$= a + \frac{2b-2a}{3} \quad (9)$$

$$= \frac{a+2b}{3}$$

$$x_3 = a + 3h$$

$$= a + \frac{3b-3a}{3}$$

$$= b$$

With the help from MATLAB, the unknown vector $\vec{a}$ (shown in Equation 7) can be solved for symbolically.

$$\begin{aligned} I &= \int_a^b f(x) dx \\ &= \int_a^b f_3(x) dx \\ &= (b-a) \times \frac{\{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\}}{8} \quad (11) \end{aligned}$$

Since

$$h = \frac{b-a}{3}$$

$$b-a = 3h$$

and Equation (11) becomes

$$I \approx \frac{3h}{8} \times \{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\} \quad (12)$$

Note the 3/8 in the formula, and hence the name of method as the Simpson's 3/8 rule.

Go through the lecture note PDF to know about the other approaches of deriving the same formula.

Simpson's 3/8 Rule

The same example

The vertical distance in meters covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$s = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

Use Simpson 3/8 rule to find the approximate value of the integral.

Simpson's 3/8 Rule

The same example

Solution

$$h = \frac{b-a}{n} = \frac{b-a}{3} = \frac{30-8}{3} = 7.3333$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$I \approx \frac{3h}{8} \times \{f(t_0) + 3f(t_1) + 3f(t_2) + f(t_3)\}$$

$$t_0 = 8$$

$$\begin{aligned} f(t_0) &= 2000 \ln \left(\frac{140000}{140000 - 2100 \times 8} \right) - 9.8 \times 8 \\ &= 177.2667 \end{aligned}$$

$$\begin{aligned} t_1 &= t_0 + h \\ &= 8 + 7.3333 \\ &= 15.3333 \end{aligned}$$

$$\begin{aligned} f(t_1) &= 2000 \ln \left(\frac{140000}{140000 - 2100 \times 15.3333} \right) - 9.8 \times 15.3333 \\ &= 372.4629 \end{aligned}$$

$$\begin{aligned} t_2 &= t_0 + 2h \\ &= 8 + 2(7.3333) \\ &= 22.6666 \end{aligned}$$

$$\begin{aligned} f(t_2) &= 2000 \ln \left(\frac{140000}{140000 - 2100 \times 22.6666} \right) - 9.8 \times 22.6666 \\ &= 608.8976 \end{aligned}$$

$$\begin{aligned} t_3 &= t_0 + 3h \\ &= 8 + 3(7.3333) \\ &= 30 \end{aligned}$$

$$\begin{aligned} f(t_3) &= 2000 \ln \left(\frac{140000}{140000 - 2100 \times 30} \right) - 9.8 \times 30 \\ &= 901.6740 \end{aligned}$$

Applying Equation (12), one has

$$\begin{aligned} I &= \frac{3}{8} \times 7.3333 \times \{177.2667 + 3 \times 372.4629 + 3 \times 608.8976 + 901.6740\} \\ &= 11063.3104m \end{aligned}$$

$$I_{\text{exact}} = 11061.34m$$

Multi-segment Simpson's 3/8 Rule

The composite version

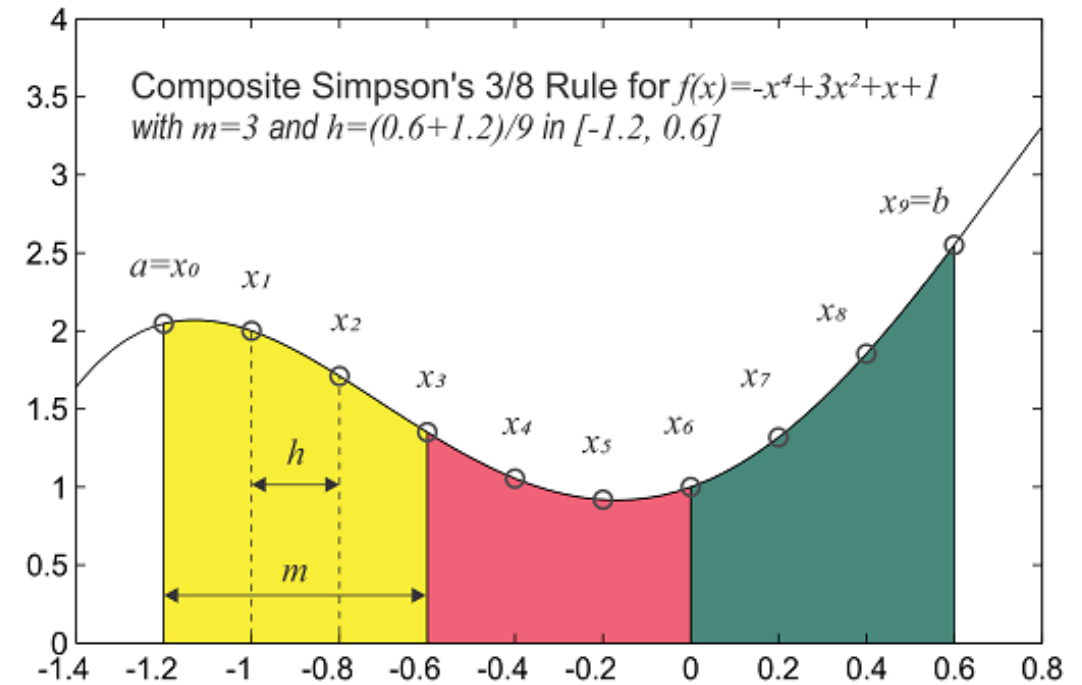
Using n = number of equal segments, the width h can be defined as

$$h = \frac{b - a}{n}$$

The number of segments need to be an integer multiple of 3 as a single application of Simpson 3/8 rule requires 3 segments.

The integral shown in Equation (1) can be expressed as

$$\begin{aligned} I &= \int_a^b f(x) dx \\ &\approx \int_a^b f_3(x) dx \\ &\approx \int_{x_0=a}^{x_3} f_3(x) dx + \int_{x_3}^{x_6} f_3(x) dx + \dots + \int_{x_{n-3}}^{x_n=b} f_3(x) dx \quad (15) \end{aligned}$$



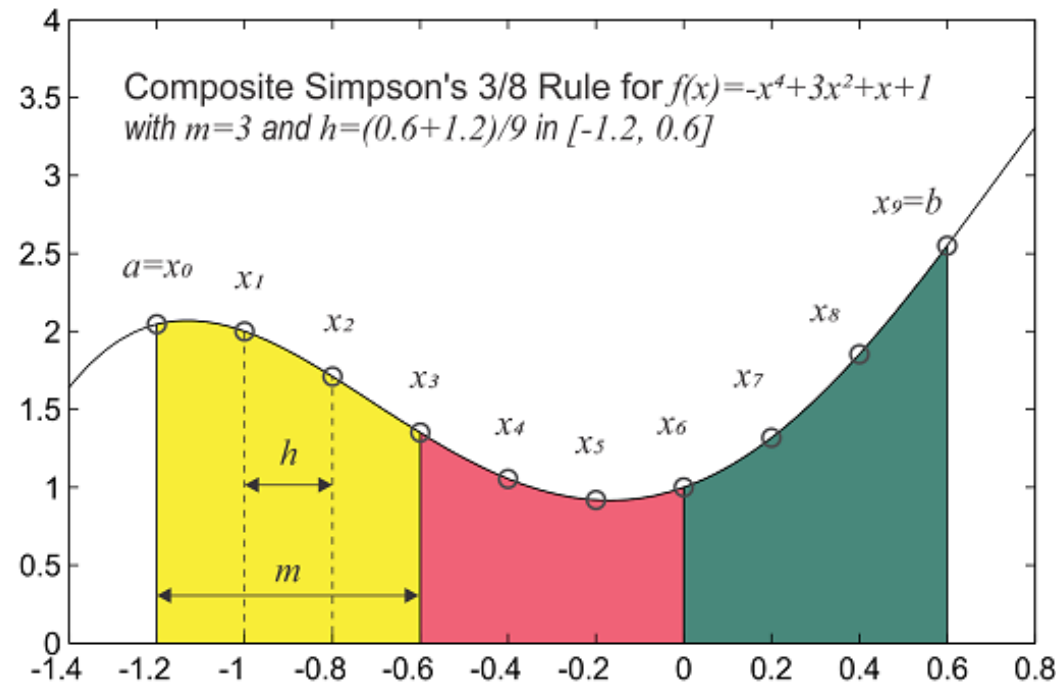
Multi-segment Simpson's 3/8 Rule

Deriving the formula

Using Simpson 3/8 rule (See Equation 12) into Equation (15), one gets

$$I = \frac{3h}{8} \left\{ f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3) + f(x_3) + 3f(x_4) + 3f(x_5) + f(x_6) \right. \\ \left. + \dots + f(x_{n-3}) + 3f(x_{n-2}) + 3f(x_{n-1}) + f(x_n) \right\} \quad (16)$$

$$= \frac{3h}{8} \left\{ f(x_0) + 3 \sum_{i=1,4,7,\dots}^{n-2} f(x_i) + 3 \sum_{i=2,5,8,\dots}^{n-1} f(x_i) + 2 \sum_{i=3,6,9,\dots}^{n-3} f(x_i) + f(x_n) \right\} \quad (17)$$



Multi-segment Simpson's 3/8 Rule

The same example (now with *multiple segments*)

The vertical distance in meters covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$s = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

Use composite Simpson 3/8 rule with six segments to estimate the vertical distance.

Multi-segment Simpson's 3/8 Rule

The same example (now with *multiple segments*)

Solution

In this example, one has (see Equation 14):

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$h = \frac{30 - 8}{6} = 3.6666$$

$$\{t_0, f(t_0)\} = \{8, 177.2667\}$$

$$\{t_1, f(t_1)\} = \{11.6666, 270.4104\} \text{ where } t_1 = t_0 + h = 8 + 3.6666 = 11.6666$$

$$\{t_2, f(t_2)\} = \{15.3333, 372.4629\} \text{ where } t_2 = t_0 + 2h = 15.3333$$

$$\{t_3, f(t_3)\} = \{19, 484.7455\} \text{ where } t_3 = t_0 + 3h = 19$$

$$\{t_4, f(t_4)\} = \{22.6666, 608.8976\} \text{ where } t_4 = t_0 + 4h = 22.6666$$

$$\{t_5, f(t_5)\} = \{26.3333, 746.9870\} \text{ where } t_5 = t_0 + 5h = 26.3333$$

$$\{t_6, f(t_6)\} = \{30, 901.6740\} \text{ where } t_6 = t_0 + 6h = 30$$

Applying Equation (17), one obtains:

$$\begin{aligned} I &= \frac{3}{8}(3.6666) \left\{ 177.2667 + 3 \sum_{i=1,4,..}^{n-2=4} f(t_i) + 3 \sum_{i=2,5,..}^{n-1=5} f(t_i) + 2 \sum_{i=3,6,..}^{n-3=3} f(t_i) + 901.6740 \right\} \\ &= (1.3750) \left\{ \begin{array}{l} 177.2667 + 3(270.4104 + 608.8976) \\ + 3(372.4629 + 746.9870) + 2(484.7455) + 901.6740 \end{array} \right\} \\ &= 11601.4696m \end{aligned}$$

Multi-segment Mixed Simpson's Rule

Combining the 1/3 and 3/8 rule

Based on the earlier discussion on (single and composite) Simpson 1/3 and 3/8 rules, the following “pseudo” step-by-step mixed Simpson rules for estimating

$$I = \int_a^b f(x)dx$$

can be given as

Step 1

n_1 = number of segments in conjunction with Simpson 1/3 rule (a multiple of 2)

n_2 = number of segments in conjunction with Simpson 3/8 rule (a multiple of 3)

Step 2

Compute

$$n = n_1 + n_2$$

$$h = \frac{b - a}{n}$$

$$x_0 = a$$

$$x_1 = a + 1h$$

$$x_2 = a + 2h$$

$$\vdots$$

$$x_i = a + ih$$

$$\vdots$$

$$x_n = a + nh = b$$

Step 3

Compute result from composite Simpson 1/3 rule

$$I_1 = \left(\frac{h}{3}\right) \left\{ f(x_0) + 4 \sum_{i=1,3,\dots}^{n_1-1} f(x_i) + 2 \sum_{i=2,4,6,\dots}^{n_1-2} f(x_i) + f(x_{n_1}) \right\}$$

Step 4

Compute result from composite Simpson 3/8 rule

$$I_2 = \left(\frac{3h}{8}\right) \left\{ f(x_0) + 3 \sum_{i=1,4,7,\dots}^{n_2-2} f(x_i) + 3 \sum_{i=2,5,8,\dots}^{n_2-1} f(x_i) + 2 \sum_{i=3,6,9,\dots}^{n_2-3} f(x_i) + f(x_{n_2}) \right\}$$

Step 5

$$I \approx I_1 + I_2$$

the final approximated answer for I .

Multi-segment Mixed Simpson's Rule

The same example (now with *mixed multiple segments*)

Compute

$$I = \int_8^{30} \left\{ 2000 \ln \left(\frac{140000}{140000 - 2100t} \right) - 9.8t \right\} dt,$$

using Simpson 1/3 rule (with $n_1 = 4$), and Simpson 3/8 rule (with $n_2 = 3$).

Multi-segment Mixed Simpson's Rule

The same example (now with *mixed multiple segments*)

Solution

The segment width is

$$h = \frac{b-a}{n} = \frac{b-a}{n_1+n_2} = \frac{30-8}{(4+3)} = 3.1429$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$t_0 = a = 8$$

$$t_1 = x_0 + 1h = 8 + 3.1429 = 11.1429$$

$$t_2 = t_0 + 2h = 8 + 2(3.1429) = 14.2857$$

$$t_3 = t_0 + 3h = 8 + 3(3.1429) = 17.4286$$

$$t_4 = t_0 + 4h = 8 + 4(3.1429) = 20.5714$$

$$t_5 = t_0 + 5h = 8 + 5(3.1429) = 23.7143$$

$$t_6 = t_0 + 6h = 8 + 6(3.1429) = 26.8571$$

$$t_7 = t_0 + 7h = 8 + 7(3.1429) = 30$$

Simpson's 1/3rule

Simpson's 3/8rule

$$\begin{aligned} f(t_0 = 8) &= 2000 \ln \left(\frac{140,000}{140,000 - 2100 \times 8} \right) - 9.8 \times 8 \\ &= 177.2667 \end{aligned}$$

$$f(t_1) = 256.5863$$

$$f(t_2) = 342.3241$$

$$f(t_3) = 435.2749$$

$$f(t_4) = 536.3909$$

$$f(t_5) = 646.8260$$

$$f(t_6) = 767.9978$$

$$f(t_7) = 901.6740$$

Multi-segment Mixed Simpson's Rule

The same example (now with *mixed multiple segments*)

For several segments ($n_1 =$ first 4 segments), using composite Simpson 1/3 rule, one obtains

$$\begin{aligned} I_1 &= \left(\frac{h}{3}\right) \left\{ f(t_0) + 4 \sum_{i=1,3,\dots}^{n_1-1=3} f(t_i) + 2 \sum_{i=2,\dots}^{n_1-2=2} f(t_i) + f(t_{n_1}) \right\} \\ &= \left(\frac{h}{3}\right) \{f(t_0) + 4(f(t_1) + f(t_3)) + 2f(t_2) + f(t_4)\} \\ &= \left(\frac{3.1429}{3}\right) \{177.2667 + 4(256.5863 + 435.2749) + 2(342.3241) + 536.3909\} \\ &= 4364.1197 \end{aligned}$$

For several segments ($n_2 =$ last 3 segments), using single application Simpson 3/8 rule, one obtains

$$\begin{aligned} I_2 &= \left(\frac{3h}{8}\right) \left\{ f(t_0) + 3 \sum_{i=1,3,\dots}^{n_2-2=1} f(t_i) + 3 \sum_{i=2,\dots}^{n_2-1=2} f(t_i) + 2 \sum_{i=3,6,\dots}^{n_2-3=0} f(t_i) + f(t_{n_1}) \right\} \\ &= \left(\frac{3h}{8}\right) \{f(t_0) + 3f(t_1) + 3f(t_2) + 2(\text{no contribution}) + f(t_3)\} \\ &= \left(\frac{3h}{8}\right) \{f(t_4) + 3f(t_5) + 3f(t_6) + f(t_7)\} \\ &= \left(\frac{3}{8} \times 3.1429\right) \{536.3909 + 3(646.8260) + 3(767.9978) + 901.6740\} \\ &= 6697.3663 \end{aligned}$$

The mixed (combined) Simpson 1/3 and 3/8 rules give

$$\begin{aligned} I &= I_1 + I_2 \\ &= 4364.1197 + 6697.3663 \\ &= 11061m \end{aligned}$$